

ESP32 / FreeRTOS  
X25519 + native ML-KEM-768  
mbedTLS AES-GCM

Wi-Fi + HTTP/TCP  
PSK-authenticated exchange  
Encrypted telemetry

Python / aiohttp  
cryptography + native ML-KEM  
In-memory session state

Research testbed: public evaluation PSK, shared directional traffic key, no telemetry AAD.